



Smart Barangay: An AI-Powered Web and Mobile Services

Portal for Barangay Tandang Sora, Butuan City

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by

Donnel L. Simbajon

Chistopher B. Raper

John Reanne P. Viray

Justin Kim P. Nadela

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CHAPTER 1 THE PROBLEM AND REVIEW OF LITERATURE

INTRODUCTION

This chapter introduces the Smart Barangay Portal, an AI-powered web and mobile system developed for Barangay Tandang Sora, Butuan City. It outlines the background of the study, review of related literature, statement of the problem and objectives, theoretical, conceptual, and research paradigm, scope and limitations, and definition of terms.

1.1 Background of the Study

Governments worldwide are increasingly adopting Artificial Intelligence (AI) as part of e-governance, using computers, mobile devices, and the internet so that systems can answer questions, process requests, and assist citizens at any time without requiring an in-person visit or human staff (Wirtz, Weyerer, & Geyer, 2019; Al-Ansi et al., 2024; United Nations, 2024; World Bank, 2022).

Al-Ansi et al. (2024) found that governments adopting such tools reported measurable gains in response speed and accuracy. In the Philippines,



this movement takes shape as the "E-Barangay" framework, encouraging barangays, the smallest government units to modernize services digitally (Bringula et al., 2022; Department of Information and Communications Technology [DICT], 2023). At the national level, DICT-led programs apply AI to improve service delivery, streamline administrative processes, and strengthen data-driven, citizen-centered governance (DICT, 2024) advancing the country's push for an efficient, transparent government that supports digital-age innovation and growth (National Economic and Development Authority [NEDA], 2023; United Nations, 2024; World Bank, 2022).

Despite these efforts, local adoption lags: as of 2024, only 318 of the country's more than 1,600 local government units (LGUs) had implemented DICT's eLGU barangay clearance system (DAP, 2024), and the Department of the Interior and Local Government's (DILG) Barangay Information Governance System (BIGS) has only recently reached the grassroots level (OpenGov Asia, 2025). Many barangays including Barangay Tandang Sora (Barangay 12) in Butuan City have yet to embrace this shift; even where digital tools exist, such as the cloud-based systems Olipas (2020) describes, they function only as basic filing systems, requiring staff to manually encode, monitor, and retrieve



records rather than answer resident questions, process requests independently, or reduce in-person visits.

In Barangay Tandang Sora, this gap is evident in daily operations. Interviews with barangay officials (Annex A) confirmed that records and services are still managed through paper logbooks, handwritten forms, and manual filing. Senior citizens, for instance, are recorded entirely by hand in notebooks vulnerable to human error, damage, and unauthorized access. Residents seeking basic information, such as clearance requirements, rely on informal inquiries or the barangay's social media page and group chats, where it is quickly buried under newer posts and hard to locate, leading to recurring misinformation, repeated inquiries, and continued reliance on in-person visits for information that could otherwise be self-served.

Residents therefore default to visiting the barangay hall in person for documents, questions, or appointments, creating long queues at peak hours; this burdens residents, who take time off work or school for routine inquiries, and staff, who spend much of the day on repeat questions instead of higher-value work. The pattern reflects a gap between current practice and two policy pushes toward digital governance: DICT's SPARK Program (Philippine Information Agency [PIA], 2024) and Butuan City's Smart Eco City vision,



which promotes technology-driven governance, innovation, and digital resilience in public service delivery (City Government of Butuan, 2024; PIA, 2025).

AI directly answers these problems: where information gets buried in group chats, AI-powered systems can give residents instant, on-demand access to it; where staff lose hours to repeat questions, chatbots can absorb routine inquiries and provide 24/7 assistance without added personnel (Wirtz, Weyerer, & Geyer, 2019; Al-Ansi et al., 2024; OECD, 2023). Evidence from other governments shows this shift raises efficiency, shortens response times, and improves citizen satisfaction (Wirtz, Weyerer, & Geyer, 2019; Al-Ansi et al., 2024; United Nations, 2024; OECD, 2023), gains that could equally modernize barangay operations toward more responsive, citizen-centered governance.

Recent advances go further: memory-augmented systems let AI retain prior interactions and generate more relevant, continuous responses, cutting repetitive input an advantage where citizens return with related or follow-up questions (Lewis et al., 2020; Borgeaud et al., 2022). Context-aware computing adds a further layer, adapting responses to user intent and interaction history to improve efficiency and user experience (Dey, 2001); applied to barangay operations, these technologies can streamline inquiries, lighten staff workload,



and give residents faster, more consistent information.

To address these problems, this study proposes the Smart Barangay Portal, an AI-integrated system with two components: a Web Portal for barangay staff and administrators, and a Mobile Application for residents.

1.2 Review of Related Literature

1.2.1 Barangay and Local Government Information Systems

Barangays, as the smallest administrative units in the Philippines, often struggle with inefficient manual processes and fragmented data management systems that hinder effective governance and timely service delivery (Tosper Jr. et al., 2025; Reyes et al., 2021). Tosper Jr. et al. (2025) addressed these challenges by developing WebYu, a web-based Barangay Information

Management System for Barangay Yuson in Guimba, Nueva Ecija. Using the Scrum development methodology and evaluating the system through the ISO/IEC 25010 software quality model, they reported an overall quality rating of “Excellent,” with particularly high scores in maintainability and security (Tosper Jr. et al., 2025). Field testing further showed substantially faster processing times for barangay certifications and fewer documentation errors



compared to manual methods, demonstrating that digitalization can significantly improve efficiency, accessibility, and transparency at the barangay level (Tosper Jr. et al., 2025; Reyes et al., 2021).

In a similar initiative, Enaño et al. (2025) developed BarangayConnect, a web-based information portal for resident data, administrative services, and community records in Barangay San Miguel, Pasig City. The system centralizes resident information, automates document requests (e.g., clearances and certificates of residency), and provides dashboards for data-driven decision-making (Enaño et al., 2025; Asian Development Bank, 2021). Evaluated using ISO/IEC 25010, BarangayConnect achieved strong ratings in efficiency, usability, and portability, though the authors recommended further enhancements in functionality and reliability (Enaño et al., 2025). Together, WebYu and similar e-governance systems illustrate that web-based barangay systems can reduce manual paperwork, speed up service delivery, and improve

transparency goals that are directly aligned with SMART BARANGAY (Asian Development Bank, 2021).

Despite these advances, existing barangay systems such as WebYu and BarangayConnect still rely primarily on structured forms and menu-driven interfaces (Tosper Jr. et al., 2025; Enaño et al., 2025). They do not incorporate advanced AI or large language model (LLM) capabilities to handle complex,



natural-language inquiries about barangay ordinances, documentary requirements, or service procedures. This limitation suggests that while digital transformation at the barangay level has progressed, there remains a gap in intelligent, conversational interaction that SMART BARANGAY aims to address.

1.2.2 E-Government and Digital Public Services

The broader literature on e-government supports the move toward digital public services as a means to improve government–citizen interaction, efficiency, and transparency. In the context of Philippine local government, Santos (as cited in Enaño et al., 2025) observed that digital platforms are increasingly replacing manual, paper-based processes for certificates, permits, and clearances, with the COVID-19 pandemic accelerating the need for contactless and efficient service delivery. However, challenges such as unstable

internet connections, limited technological infrastructure, and gaps in digital literacy among staff still hinder full implementation, highlighting the importance of user-friendly, reliable systems and continuous capacity-building (Enaño et al., 2025).

At the global level, the United Nations (2022) emphasized in its E-Government Survey that digital government initiatives should prioritize



inclusivity, accessibility, and responsiveness to citizen needs, especially in local government units that serve as the first point of contact for many residents. Digital public services improve government–resident interaction by enabling remote access to transactions, reducing waiting times, and increasing transparency through traceable processes (United Nations, 2022). Yet many e-government platforms still focus primarily on transactional efficiency without integrating intelligent assistance that can guide citizens through complex procedures or answer diverse questions in natural language. SMART BARANGAY extends this e-government vision by combining digital service automation with AI-powered conversational support, thereby enhancing both accessibility and user experience at the barangay level.

1.2.3 Web-Based and Mobile-Based Information Systems

Web-based and mobile-based information systems are central to modernizing barangay services by enabling residents to access transactions and information through multiple devices and platforms. Enaño et al. (2025) designed BarangayConnect using standard web technologies (HTML, CSS, JavaScript, PHP, and MySQL), creating a responsive and user-friendly interface



accessible on both computers and smartphones via browsers. The system supports automated request handling, record management, and report generation through an intuitive dashboard for barangay staff, demonstrating that web-based platforms can effectively centralize resident data and streamline administrative services (Enaño et al., 2025).

The integration of web and mobile access is particularly important for barangay services because many residents rely on smartphones as their primary device for communication and information access. BarangayConnect is accessible via mobile browsers but does not include a dedicated mobile application, and the authors explicitly recommend a separate mobile app to improve accessibility and user experience (Enaño et al., 2025). SMART BARANGAY responds to this need by providing both a web portal for staff and administrators and a dedicated mobile application for residents, ensuring synchronized data and consistent experiences across platforms.

Integrated web-mobile platforms typically rely on API-based communication to synchronize data between front-end applications and the back-end database, ensuring that requests, records, and status updates remain consistent regardless of the device used. This architecture supports real-time monitoring of service requests, document tracking, and seamless



handover between mobile and web interfaces, which are essential features for SMART BARANGAY's request tracking and monitoring capabilities.

1.2.4 Artificial Intelligence in Government Services

Artificial intelligence (AI) is increasingly being adopted in public administration to support decision-making, automate routine tasks, and enhance service delivery. Singh and Hess (2026) examined AI chatbots in public services across several countries and found that AI-based conversational agents have significant potential to improve efficiency, citizen engagement, and accessibility, particularly for structured query services such as basic information requests and status checks. Their findings show that AI chatbots can operate as first-line support, providing 24/7 access to information and freeing human staff to focus on more complex tasks (Singh & Hess, 2026).

However, AI implementation in government also presents challenges. AI systems depend on training data that may be incomplete or biased, and in public services, inaccurate or inappropriate outputs can undermine trust and accountability (Singh & Hess, 2026). To mitigate these risks, public-sector AI systems must rely on curated knowledge sources, transparent decision logic,



and mechanisms for human oversight, particularly when dealing with sensitive or high-impact decisions.

Intelligent government systems that combine AI with structured workflows can automate routine inquiries, guide citizens through procedures, and escalate more complex cases to human personnel. This aligns with SMART BARANGAY's goal of using AI to handle repetitive questions and service requests, thereby freeing barangay staff to focus on tasks requiring human judgment. Unlike general-purpose chatbots, SMART BARANGAY's AI will be explicitly constrained to barangay policies and records, helping ensure that responses are accurate and aligned with local governance rules.

1.2.5 Large Language Models (LLMs) and Generative AI

Large language models (LLMs), including GPT-based systems, have advanced natural language understanding and generation capabilities that

enable them to answer complex questions, explain procedures, and generate human-like text. In government contexts, LLMs can be applied to support citizen inquiries, document drafting, and policy explanation, provided that their outputs are grounded in reliable, domain-specific information. Retrieval-Augmented Generation (RAG) techniques enhance LLMs by



combining them with external knowledge bases, so that the model retrieves relevant information from curated documents (such as ordinances and service guidelines) before generating a response.

In barangay services, LLMs can support residents by answering questions about documentary requirements, explaining step-by-step processes for obtaining clearances, and clarifying policy details in plain language. Compared with traditional rule-based systems, LLMs can handle diverse phrasings, understand follow-up questions, and maintain context across multiple conversational turns. At the same time, LLMs carry risks of “hallucinating” unsupported information if not properly constrained, which is not acceptable for official government advice.

SMART BARANGAY addresses these concerns by integrating an LLM chatbot that is grounded in the barangay’s official rules, policies, and service guidelines. By combining LLM capabilities with RAG and a controlled

knowledge base, the system aims to provide flexible yet trustworthy responses tailored to barangay-specific information needs.

1.2.6 AI Chatbots and Conversational Systems



AI chatbots and conversational systems are increasingly used as virtual assistants in e-government service platforms. In a study on intelligent chatbots as frontline agents in e-government service platforms, researchers reported that chatbots can significantly reduce response time, improve consistency of information, and enhance citizen satisfaction when properly designed and integrated into service workflows (Intelligent Chatbots as Frontline Agents, n.d.). Chatbots can handle frequently asked questions, guide users through procedures, and collect initial data before handing off to human staff, thereby acting as a scalable interface between citizens and government services.

Rule-based chatbots, which rely on predefined intents and fixed response templates, are limited in their ability to handle varied natural-language inputs and complex conversational flows. They often fail when users phrase questions unexpectedly or ask follow-up questions outside predefined paths, which can lead to user frustration and reduced trust in digital services. LLM-powered chatbots, by contrast, are better able to interpret diverse phrasing and

maintain context over multiple turns, making them more suitable for complex interactions such as multi-step applications or clarifications of local regulations.

However, LLM-powered chatbots in government settings require strict control to ensure that responses remain accurate, appropriate, and aligned



with official policies. SMART BARANGAY incorporates strategies such as restricting the chatbot to a curated knowledge base, using RAG, and enforcing session-based privacy controls. The chatbot is trained specifically on barangay rules and policies, ensuring that responses are grounded in official documentation rather than open internet sources.

1.2.7 Knowledge Management and Information Retrieval

Knowledge management and information retrieval are critical for enabling residents to access barangay ordinances, requirements, announcements, and service guidelines efficiently. AI-powered information access systems combine structured knowledge bases with semantic search mechanisms to allow users to pose queries in natural language and receive relevant results. In government settings, such systems enable citizens to locate specific policies, understand

procedural steps, and identify necessary documents without navigating complex websites or static documents.

BarangayConnect illustrates the value of integrating analytical and knowledge-related capabilities within a barangay system. Enaño et al. (2025) incorporated data analytics and linear regression algorithms to analyze



demographic and community data, supporting data-driven governance and service planning. While BarangayConnect focuses more on analytics than on conversational retrieval of text-based knowledge, it demonstrates that barangay-level systems can support more advanced functions than simple record-keeping (Enaño et al., 2025). SMART BARANGAY extends this vision by incorporating a knowledge-based chatbot that retrieves and explains information from official barangay documents, ordinances, and service guidelines.

By centralizing official rules and policies in a structured knowledge base and linking them to the LLM chatbot, SMART BARANGAY ensures that residents can ask questions such as “What do I need to get a barangay clearance?” and receive accurate, policy-grounded answers. This approach improves information accessibility and reduces reliance on informal inquiries or unorganized social media posts, which may be incomplete or outdated.

1.2.8 Workflow Automation and Digital Request Management

Workflow automation and digital request management are essential for modernizing barangay administrative processes such as the issuance of clearances, certificates, and other resident documents. Digital



request systems allow residents to submit applications online, track their status, and receive notifications when documents are ready, thereby reducing physical visits and queuing at the barangay hall. These systems also create clearer audit trails and improve transparency by recording each step of the process.

WebYu demonstrated that digitizing barangay processes can substantially reduce processing time and documentation errors, with field testing showing significantly faster handling of certifications and fewer manual mistakes compared to traditional approaches (Tosper Jr. et al., 2025). BarangayConnect similarly automates the issuance of documents such as clearances and certificates of residency, reducing manual work while providing dashboards that allow officials to monitor workloads and service performance (Enaño et al., 2025).

SMART BARANGAY extends these capabilities by tightly integrating workflow automation with its AI chatbot. When a resident describes a need—such as

requesting a barangay clearance—the chatbot can identify the relevant service, present the appropriate digital form, and guide the resident through



the required fields within the conversation. After submission, the system enables both residents and staff to track request status, ensuring timely processing and reducing uncertainty about where an application stands in the workflow.

1.2.9 AI Memory and Context-Aware Systems

AI memory and context-aware systems allow chatbots to maintain conversation history and interpret user input based on prior exchanges, leading to more natural and efficient interactions. Session-based AI memory enables the chatbot to recall previous messages within a session, understand follow-up questions, and provide coherent, context-sensitive responses. This capability is especially important for multi-step processes such as document applications, where residents may need to ask sequential questions or clarify prior information.

In SMART BARANGAY, AI memory is used to improve the resident experience by allowing the chatbot to remember key details within a session, such as the type of document the resident is applying for or the specific requirements already discussed. This reduces the need for residents to repeat information



and supports smoother, more conversational guidance through procedures. At the same time, privacy considerations are paramount. The Philippine Data Privacy Act of 2012 (Republic Act No. 10173) requires that personal data be handled securely and only for legitimate, authorized purposes. SMART BARANGAY complies with this by ensuring that each chatbot session is private and isolated, with no access by other users or staff, and that session data is deleted after use in accordance with documented policies.

1.3 Theoretical Framework

1.3.1 DeLone and McLean Information Systems Success Model

DeLone and McLean (2003) proposed a comprehensive model for measuring the success of information systems through six interdependent dimensions: System Quality, Information Quality, Service Quality, Use, User Satisfaction, and Net Benefits. Originally introduced in 1992 and updated in 2003 to incorporate service quality as a distinct evaluative dimension, this model has been extensively applied in e-government research to assess digital public service platforms. Its utility lies in its capacity to connect technical performance characteristics with user-centered outcomes, providing a holistic



framework for evaluating both the operational and experiential success of information systems in real-world deployment contexts.

In the context of the Smart Barangay Portal, the DeLone and McLean model directly informs how the system's success is conceptualized and measured across its web portal, mobile Progressive Web Application (PWA), and AI chatbot components. System Quality is reflected in the platform's responsiveness, processing speed, and cross-device compatibility attributes addressed through Bootstrap 5 responsive design and cloud-hosted Supabase infrastructure. Information Quality is embodied in the AI chatbot's capacity to deliver accurate, barangay-specific responses grounded in officially uploaded ordinances, service guidelines, and administrative policies. Service Quality encompasses the system's 24/7 availability, notification delivery reliability, and service request processing efficiency. User Satisfaction is directly assessed through the study's structured User Acceptance Testing (UAT) instrument. Net Benefits are realized through the projected reduction in in-person barangay visits, decreased administrative workload, and improved resident access to government services outcomes that correspond directly to the system's primary design objectives (DeLone & McLean, 2003).

1.3.2 Human-Computer Interaction Theory

Human-Computer Interaction (HCI) is an interdisciplinary field that examines the design, evaluation, and implementation of interactive computing systems for human use, with a central focus on usability, user experience, and accessibility (Dix, Finlay, Abowd, & Beale, 2004). Nielsen's (1994) foundational usability heuristics including learnability, efficiency, memorability, error prevention, and satisfaction provide concrete evaluative criteria for assessing the quality of digital interfaces. A core principle of HCI is that effective system design must accommodate the diverse characteristics, abilities, and expectations of intended users, ensuring that technology serves as an enabler of access rather than a barrier to participation.

HCI theory shapes the design decisions of both the resident-facing and staff-facing components of the Smart Barangay Portal. For resident-users a population that encompasses youth, working adults, and senior citizens with varying levels of digital literacy the mobile PWA's chatbot interface adopts a conversational interaction model that minimizes technical barriers and reduces the need for prior technology training. The chatbot's support for queries in Bisaya, Filipino, and English reflects an HCI-informed design principle of linguistic accessibility that prioritizes natural communication. For barangay staff, the administrative web portal employs a structured dashboard layout

that organizes service request management, document uploads, and announcement publishing into clearly defined workflows, reducing cognitive load and the likelihood of operational error. These design decisions are further reinforced by the system's PWA architecture, which ensures consistent interface behavior across mobile phones, tablets, and desktop computers regardless of device capability (Dix et al., 2004; Nielsen, 1994).

1.3.3 Conversational AI and Natural Language Processing Theory

Conversational AI encompasses the design and implementation of computational systems capable of interpreting and generating human language in contextually meaningful and communicatively effective ways. The transformer-based neural architecture introduced by Vaswani et al. (2017) established the theoretical foundation for modern large language models (LLMs) by enabling systems to capture complex linguistic relationships and long-range contextual dependencies through self-attention mechanisms. Building upon this architecture, Brown et al. (2020) demonstrated that large-scale language models can perform diverse natural language processing tasks with high accuracy through in-context learning, establishing the theoretical basis for LLM integration in specialized, domain-specific application environments.



The Smart Barangay Portal's AI-powered chatbot is grounded in Conversational AI and NLP theory through its LangChain-based implementation. The chatbot employs transformer-based intent recognition to process multilingual resident queries accepting natural language inputs in Bisaya, Filipino, and English and maps these inputs to contextually appropriate informational responses or service request workflows. The session-based AI memory architecture, which maintains conversational context across multiple query exchanges within a single resident session, reflects the theoretical model of context-aware dialogue management that is central to modern conversational AI systems. This memory capability enables the chatbot to deliver coherent and continuous assistance rather than treating each user query in isolation directly supporting the study's objective of implementing session-based AI memory for personalized and continuous barangay service delivery (Vaswani et al., 2017; Brown et al., 2020).

1.3.4 A Graph-Based Retrieval-Augmented Generation Approach

Information Retrieval (IR) theory addresses the computational processes by which relevant information is identified, retrieved, and presented from



structured or unstructured document collections in response to a user's expressed information need (Manning, Raghavan, & Schutze, 2008). Classical

IR models including the vector space model and probabilistic retrieval frameworks establish the foundational logic for matching user queries to document content based on computed relevance scores. Lewis et al. (2020) extended this foundation through the Retrieval-Augmented Generation (RAG) framework, which combines neural document retrieval with generative language models to produce factually grounded, contextually accurate outputs substantially reducing the hallucination and factual inaccuracy problems associated with standalone LLMs operating without access to verified source documents.

The Smart Barangay Portal's AI chatbot is architecturally grounded in RAG theory, implemented through LangChain's document retrieval pipeline integrated with a Supabase vector database. Barangay officials upload official ordinances, service guidelines, and policy documents to the system, which are subsequently indexed as domain-specific vector embeddings representing the barangay's authoritative knowledge base. When a resident submits a query, the RAG mechanism retrieves the most semantically relevant document passages and uses these as the factual grounding for the chatbot's generated response ensuring that answers accurately reflect the barangay's official policies rather



than generalized or potentially inaccurate internet content. This design directly addresses one of the core problems identified in the study: residents

receiving outdated or incorrect barangay information through informal channels such as social media. By anchoring AI-generated responses to verified, locally uploaded documents, the system operationalizes IR and RAG theory as a practical mechanism for accurate, policy-consistent public information delivery at the barangay level (Manning et al., 2008; Lewis et al., 2020).

1.4 Conceptual Framework

The conceptual framework of this capstone study adopts the Input-Process-Output (IPO) model, which integrates the principles of Retrieval-Augmented Generation (RAG) Theory, Conversational AI and Natural Language Processing (NLP) Theory, and the Data Privacy Act of 2012 (Republic Act No. 10173). The IPO model is appropriate for this study because it reflects the design, development, and evaluation of an AI-powered web and mobile system by illustrating the flow of data from resident inquiry to administrative output.



The input component consists of resident-generated data, specifically inquiries and service requests submitted in Bisaya, Filipino, and English, as well as staff-managed data such as uploaded ordinances, service guidelines, and identity validation records used to authenticate users and administrators.

In accordance with Conversational AI and NLP Theory, these inputs must be accurately captured and interpreted across multiple languages and linguistic variations to enable the system to generate contextually relevant and policy-based responses rather than generic or unsupported replies.

The process component represents the core operations of the system. It includes access to the portal and identification of user type, natural language processing of resident queries, dynamic form generation for service requests, validation and storage of submitted records, staff review and approval or rejection of requests, and the publication of approved announcements and notifications. Guided by RAG Theory, the NLP module retrieves the most relevant text segments from the barangay's officially uploaded knowledge base before generating a response, ensuring that outputs are grounded in verified barangay policies rather than unrestricted external sources. In compliance with Republic Act No. 10173, each resident session is treated as private and isolated, with session memory cleared immediately after the interaction ends to ensure that personal data is not retained beyond its authorized purpose.

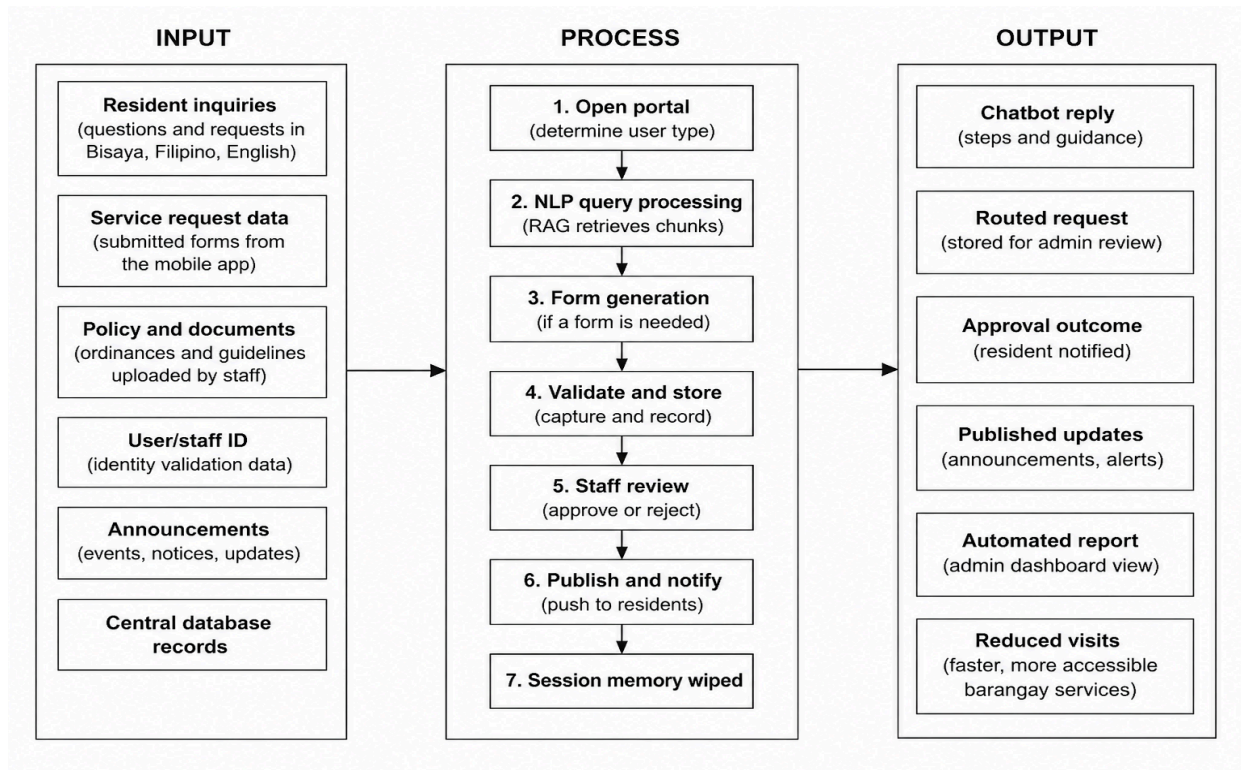


The output component refers to the results produced by the interaction between the resident and the system, as well as the administrative actions that follow. These outputs include chatbot responses that provide instructions and guidance, routed service requests for administrative review, approval or rejection notices communicated to residents, published announcements and

alerts delivered through notifications, and automated reports generated for the administrative dashboard. Beyond these immediate system outputs, the study also anticipates broader outcomes, including reduced in-person barangay visits and improved accessibility and efficiency in public service delivery. Collectively, these outputs demonstrate the system's ability to transform resident inquiries and administrative inputs into accurate, policy-consistent, and timely barangay services.

Research Paradigm

Figure 1.1 Input - Process - Output (IPO)



1.5 Statement of the Problem

Despite national efforts to promote digital local governance, adoption of systems such as the DICT's eLGU barangay clearance system and the DILG's Barangay Information Governance System remains limited (DAP, 2024; OpenGov Asia, 2025). In Barangay Tandang Sora, records and resident services are still handled through paper logbooks, handwritten forms, and manual filing, making information retrieval slow and prone to error (Annex A). Residents also rely on informal channels such as social media pages and group chats for barangay information, which can lead to outdated guidance and



repeated inquiries (Annex A). As a result, residents and staff experience delays and unnecessary time spent on routine transactions. This study addresses this gap through the development of the Smart Barangay Portal, an AI-enabled system for faster and more accessible barangay services (City Government of Butuan, 2023; DICT, 2024; NEDA, 2023; PIA, 2024).

1.6 Objectives of the Study

The main objective of this study is to develop SMART BARANGAY, an AI-powered (i.e. web and mobile) system that modernizes the administrative

and constituent services of Barangay Tandang Sora, Butuan City. In order to achieve the main objective is to achieve the following specific objectives:

- To design and develop web and mobile portals with synchronized data access, enabling residents to avail of barangay services remotely.
- To integrate a large language model (LLM) into the barangay system to accurately process and respond to resident inquiries and document requests.
- To implement administrative services in accordance with



barangay standards, encompassing the management of ordinances, service guidelines, and announcements, as well as the monitoring of resident service requests (e.g., barangay clearances, certificates of indigency, proof of income, and barangay IDs, among others). This process is supported by a session-based AI memory system that retains resident interaction context across conversations, enabling personalized and continuous service monitoring.

- To implement an AI-powered chatbot that addresses resident inquiries and provides continuous (24/7) automated assistance for barangay services.
- To automate the end-to-end service process, from the initial submission of requests to the final completion and delivery of documents.

1.7 Significance of the Study

The significance of this study lies in its potential to improve the delivery of barangay services through an AI-powered web and mobile portal developed for Barangay Tandang Sora, Butuan City. By integrating a digital request



system, an administrative dashboard, and an AI chatbot, the proposed Smart Barangay Portal may help streamline service delivery, reduce manual workloads, and improve access to accurate and timely information for residents.

This study is beneficial to the following stakeholders:

Barangay Officials and Staff.

The system may lessen the burden of repetitive inquiries and manual recordkeeping by automating routine processes such as information dissemination, request monitoring, and document handling. As a result, barangay personnel may focus more on tasks that require direct human judgment and administrative attention.

Residents of Barangay Tandang Sora.

Residents may benefit from faster and more convenient access to barangay information and services through the mobile application and AI chatbot. Instead of visiting the barangay hall for every inquiry or request, they may submit forms, ask questions, and track transactions remotely, thereby saving time and effort.



Barangay Administration and Local Governance.

The portal may support better organization of records, improved monitoring of requests, and more efficient reporting. In addition, the system may contribute to transparent and responsive governance by providing a centralized platform for service delivery and communication.

Vulnerable and Time-Constrained Users.

Senior citizens, working residents, and other users with limited time or mobility may particularly benefit from remote access to barangay services. The system's privacy controls and session-based chatbot interaction also help protect user information while providing assistance.

Future Researchers and Developers.

This study may serve as a reference for future researchers who wish to develop similar AI-powered local government systems. It may also provide a basis for further improvements in digital governance, chatbot integration, and

barangay-level service automation.



1.8 Scope and Limitation of the Study

This study focuses on the design, development, deployment, and evaluation of the Smart Barangay Portal for Barangay Tandang Sora, Butuan City. The system is composed of two main components: a web portal intended for barangay officials and administrators, and a mobile application intended for residents.

The scope of the study includes the following functionalities: uploading and managing barangay rules, policies, and service guidelines; posting announcements and updates; receiving and monitoring service requests; providing AI-powered responses to resident inquiries; and enabling residents to access forms and submit requests digitally. The study also covers the implementation of session-based AI memory, a retrieval-augmented generation chatbot, and automated request tracking within the platform.

The study is delimited to Barangay Tandang Sora and does not claim to represent all barangays in Butuan City or in the Philippines. Its findings, performance, and user feedback are limited to the selected respondents and the actual operational setting of the barangay. Thus, the results may not be directly generalized to other barangays without further testing and adaptation.



1.9 Definition of Terms

The following terms are defined operationally as used in this study:

Artificial Intelligence refers to the technology used by the Smart Barangay Portal to automate tasks, understand user inquiries, and provide intelligent assistance to residents.

Barangay Official refers to an authorized personnel who manages barangay services, records, requests, and announcements using the system.

Chatbot refers to the virtual assistant integrated into the Smart Barangay Portal that interacts with users and provides information regarding barangay services and procedures.

Document Request refers to the process of applying for barangay-issued documents such as clearances, certificates, and other official records through the system.

Information Retrieval refers to the process of locating and providing relevant information from stored documents and records based on a user's query.

Knowledge Base refers to the collection of official barangay documents, policies, ordinances, guidelines, and service information used by the chatbot.

Large Language Model refers to an artificial intelligence technology capable of understanding and generating human language for conversational interactions.



Natural Language Processing refers to the technology that enables computers to understand, interpret, and respond to human language.

Notification refers to an automated message sent by the system to inform users about updates, announcements, and request statuses.

Progressive Web Application refers to a web-based application that provides features and functionality similar to a mobile application.

Resident refers to an individual living within Barangay Tandang Sora who accesses the services provided by the Smart Barangay Portal.

Resident Inquiry refers to a question, concern, or request for information submitted by a resident through the system.

Retrieval-Augmented Generation refers to an AI approach that retrieves information from trusted documents before generating responses.

Role-Based Access Control refers to a security mechanism that restricts access to system features based on a user's assigned role and permissions.

Service Request refers to an application, transaction, or request submitted by a resident for processing by barangay personnel.

Session-Based Memory refers to the chatbot's ability to temporarily retain information during an active conversation to provide contextually relevant responses.



Smart Barangay Portal refers to the AI-powered web and mobile system developed in this study to improve barangay service delivery, communication, and information management.

User Authentication refers to the process of verifying a user's identity before granting access to the system.

Web Portal refers to the administrative component of the Smart Barangay Portal used by barangay personnel to manage services, records, requests, and announcements.

Workflow Automation refers to the use of technology to automatically perform and manage routine processes with minimal human intervention.

CHAPTER 2 METHODOLOGY

This chapter presents the research methods and procedures used in the development of **Smart Barangay**. It describes how the Smart Barangay Portal was developed and evaluated. It covers the research design, the step-by-step development process, the participants involved, the survey instrument used to measure user acceptance, the ethical standards observed, and the statistical methods applied to analyze the results.

2.1 Research Design

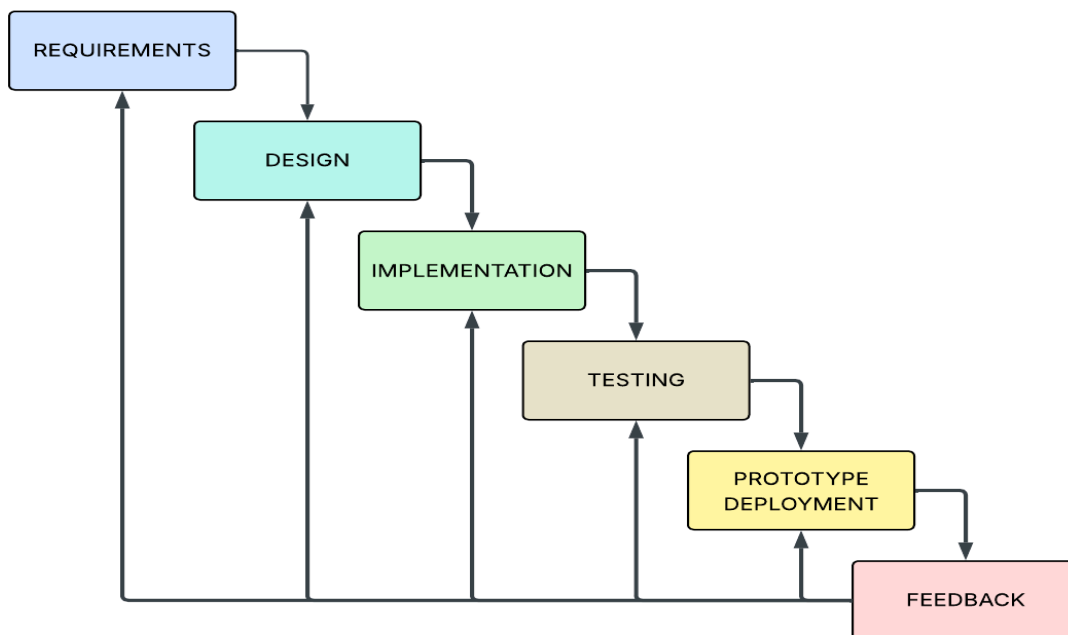
2.1.1 Software Development Model

This study utilized the Waterfall Model as its software development methodology. The Waterfall Model was a sequential and structured approach in which each phase of development was completed in full before progressing to the next, ensuring that requirements, design specifications, and compliance considerations were clearly defined and approved at the earliest stages of the project. This model was deemed appropriate for the development of the Smart Barangay Portal because the system operated within a fixed regulatory and procedural environment governed by Republic Act No. 10173 (Data Privacy Act

of 2012) and guidelines issued by the Department of the Interior and Local Government (DILG). The clearly defined and largely stable nature of these legal and administrative requirements made an upfront, linear development approach more suitable than an iterative one, as core functionalities such as data privacy safeguards, restricted access protocols for sensitive records, and resident service workflows needed to be established and validated before development proceeded to subsequent stages.

2.1.2 Phases of the Waterfall Model

Figure 2.1 WaterFall Model Framework.



It illustrates the Waterfall Model used in the development of the Smart Barangay system. It shows the sequential cycle of requirements analysis, system design, development, testing, deployment, and maintenance, which allowed the researchers to systematically develop, evaluate, and improve the Smart Barangay system throughout the project.

Table 2.1 Project Development Timeline.

Phase	Duration	Focus / Deliverables
Requirements	Weeks 1-2	Stakeholder consultation, documentation of functional and legal requirements
System Design	Weeks 3-4	Architecture planning, database schema design, and interface wireframes
Implementation	Weeks 5-7	Development of the API, RAG AI chatbot, database, and PWA/admin portal interfaces
Integration and	Weeks 8 - 9	System integration,

Testing		functional testing, and data privacy compliance verification
Deployment	Weeks 10 - 11	System release, cloud configuration, and stakeholder onboarding
Operation and Maintenance	Weeks 12	Post-deployment monitoring, issue resolution, and stakeholder evaluation

2.1.3 Requirements Analysis and Specification

The Web Portal functions as a digital office for the barangay. Authorized staff can upload official rules, policies, and service guidelines; post announcements and community updates; and receive and process service requests submitted through the mobile app. All records are stored in one secure, organized location, simplifying information management and reducing the risk of lost or damaged documents.

The Mobile Application is designed for residents and features an AI-powered chatbot—a smart virtual assistant trained specifically on the barangay’s official rules and policies. When a resident asks, “What do I need to



get a barangay clearance?”, the chatbot responds using the barangay’s officially documented information rather than random internet sources. The chatbot can also detect when a resident needs to submit a form and automatically display the correct form within the conversation, allowing the resident to fill it out and submit it on their phone without leaving home.

A critical feature of the system is resident privacy protection. Each chatbot conversation is completely private and isolated; no other user, barangay staff member, or administrator can view what a resident asked or submitted. At the end of each session, all data is permanently deleted. This design complies with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), which protects every Filipino citizen’s right to privacy in digital interactions.

By integrating an AI-powered chatbot, a secure digital records system, and an automated service request process into a single platform, the Smart Barangay Portal aims to reduce the need for in-person barangay hall visits, accelerate service request processing, and free barangay staff to focus on tasks requiring personal attention.

Requirements Analysis and Specification – This phase involved the identification of system requirements through review of barangay



administrative processes, applicable legal mandates, and consultation with barangay personnel and resident representatives. Functional requirements for the AI Chatbot (Agent), service request handling, and notification system, as well as non-functional requirements such as data privacy compliance, were documented during this stage.

System Design – This phase translated the documented requirements into the system's architectural blueprint, encompassing the design of the Barangay Services Hub, the integration of the Retrieval-Augmented Generation (RAG) AI module, the Application Programming Interface (API) structure, and the database schema. Interface design for both the administrative web portal and the resident-facing Progressive Web Application (PWA) was also finalized during this stage.

Implementation – This phase involved the actual coding and construction of the system components, including the development of the FastAPI-based API layer, the LangChain-powered RAG AI chatbot, and the Supabase database, in accordance with the specifications established in the design phase.

Integration and Testing – This phase combined the individually developed modules into a unified system and subjected the integrated platform to functional, performance, and security testing to verify that all components

operated as intended.

Deployment – This phase involved the release of the completed system to its intended environment for use by barangay staff and residents, including the configuration of cloud infrastructure through Supabase and the Node.js runtime environment.

Operation and Maintenance – This phase covered the ongoing monitoring of system performance following deployment, along with the resolution of issues identified during actual use, to ensure the continued reliability and security of the platform.

Project Team Roles

Project Leader – Oversees the overall direction of the study, defines project objectives, and ensures that each development phase aligns with the requirements of barangay stakeholders.

Systems Analyst – Responsible for gathering and documenting requirements, as well as translating barangay administrative processes into functional and technical specifications.

Developer – Responsible for the design, coding, and integration of the system's

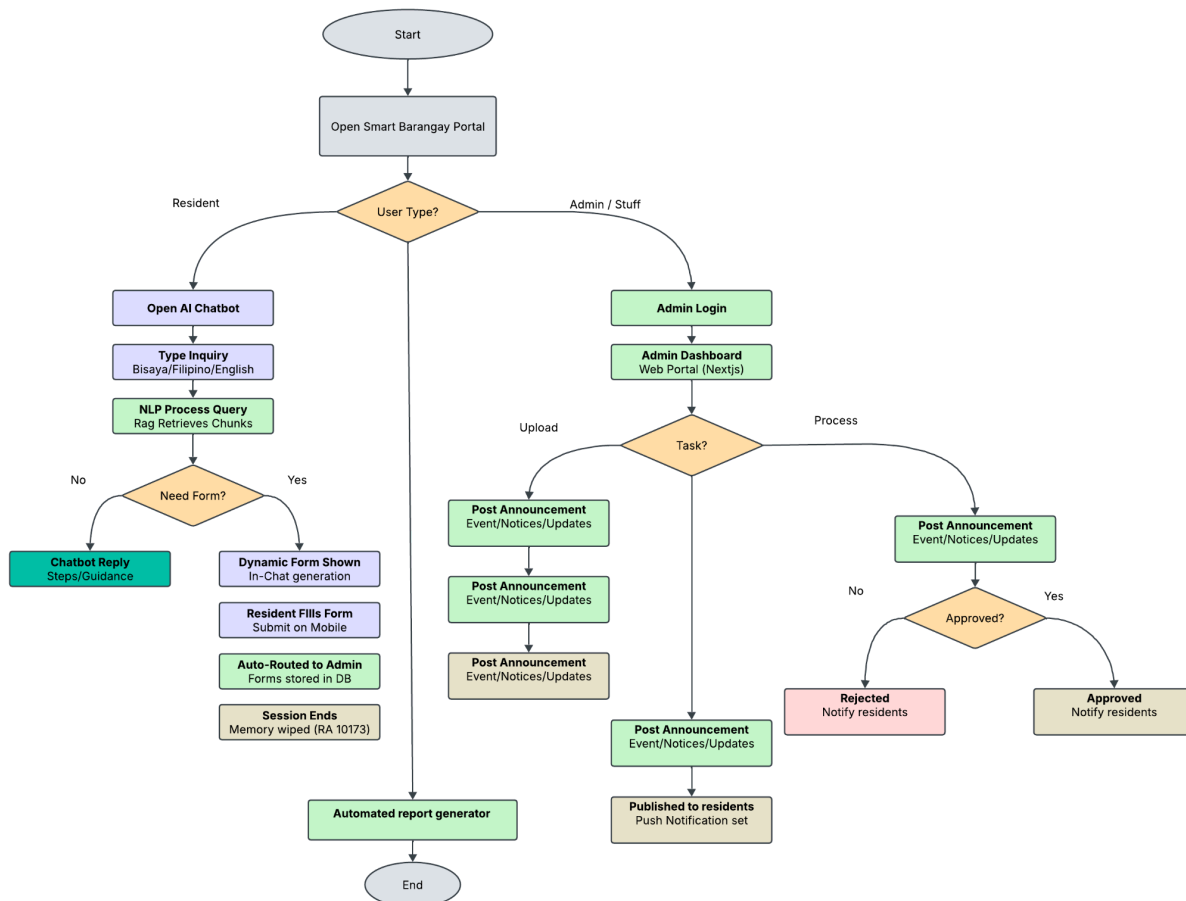


hardware-independent software components, including the AI agents, API, and database.

Quality Assurance Evaluator – Responsible for conducting system testing, validating compliance with data privacy requirements, and verifying the accuracy and reliability of system outputs prior to deployment.

System Architecture

Figure 2.2 System Flowchart.



The system flowchart illustrates the sequential logic governing user interaction within the Smart Barangay Portal, beginning with the initiation of the platform and branching according to the type of user accessing the system,



either a resident or a barangay administrator.

For resident-users, the portal directs the user to the AI-powered chatbot, where an inquiry may be entered in Bisaya, Filipino, or English. The chatbot's underlying Retrieval-Augmented Generation (RAG) mechanism applies natural language processing to interpret the query and retrieve relevant contextual information from the knowledge base. The system subsequently determines whether the inquiry requires the completion of a form. If no form is required, the chatbot generates a direct reply containing the appropriate steps or guidance. If a form is required, the system dynamically generates the corresponding form within the chat interface, which the resident completes and submits through the mobile application. The submitted form is then automatically routed to the barangay administrator and stored in the database, after which the session concludes and the session memory is cleared in compliance with Republic Act No. 10173, or the Data Privacy Act of 2012.

For barangay administrators, access begins with an administrator login, which leads to the administrative dashboard built on a Next.js web portal. From the

dashboard, the administrator selects the specific task to be carried out, which may involve posting announcements, notices, event schedules, or general



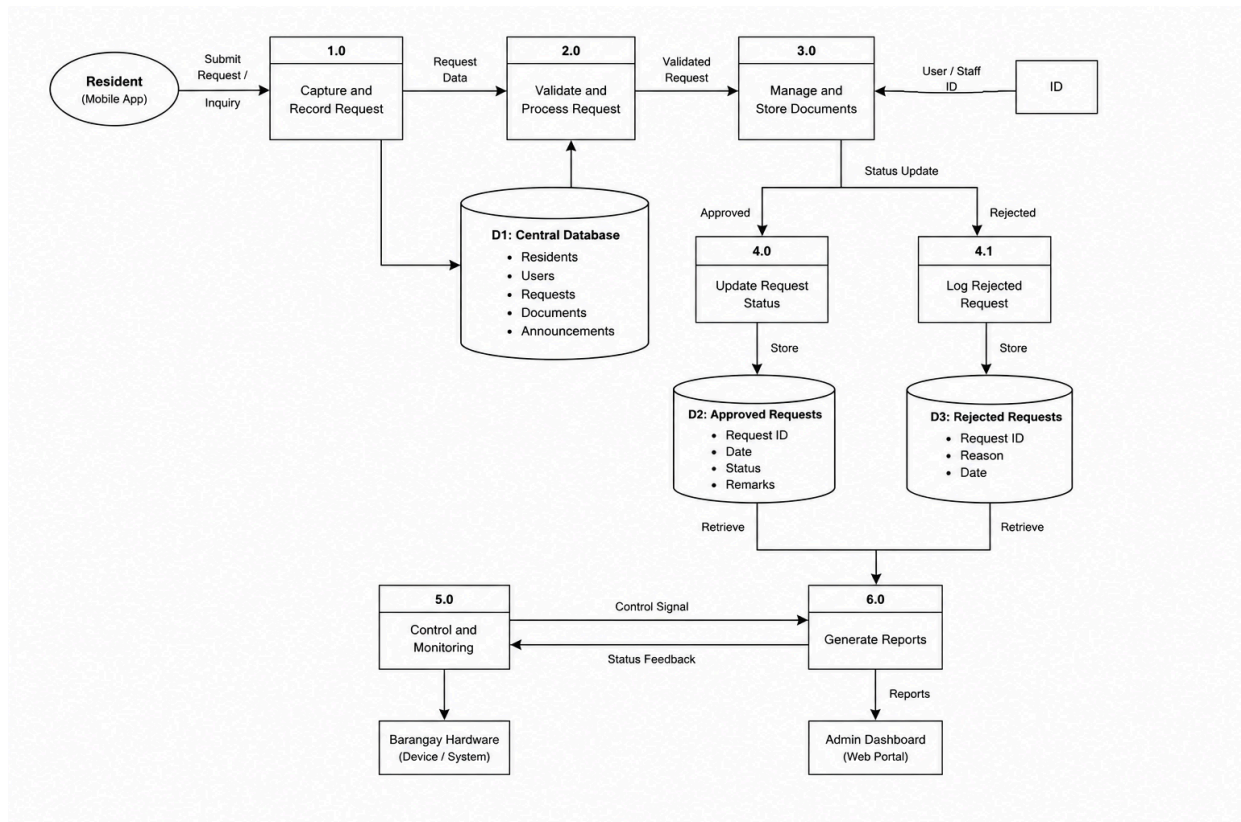
updates intended for residents, or reviewing and acting upon resident-submitted service requests. In the case of service requests, the administrator evaluates the submission and determines whether to approve or reject it; in either outcome, the resident is notified accordingly. Once an announcement or update has been finalized, it is published to residents through a push notification mechanism, ensuring the timely dissemination of barangay information.

Both the resident-facing and administrator-facing processes ultimately converge into an automated report generation function, which compiles relevant interaction and transaction data for documentation and monitoring purposes, before the process concludes.

Data Flow Diagram (DFD)

Figure 2.3 Data Flow Diagram.

The DFD illustrates how data is moved between the system components.



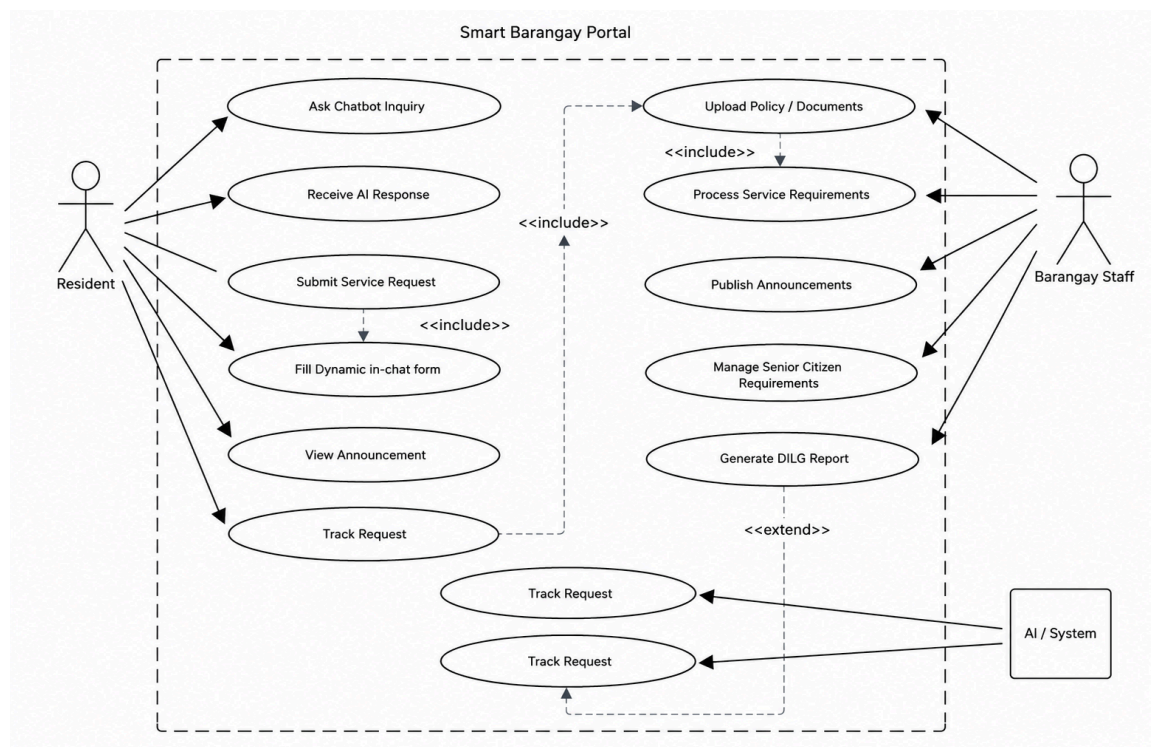
The Data Flow Diagram shows how the Smart Barangay Portal processes and manages data. Residents submit requests and inquiries through the mobile application, which are processed by the AI chatbot and barangay services. Information is stored in the database, documents are indexed

for AI retrieval, notifications are sent to residents, and reports are generated for barangay operations. This enables efficient service delivery, record management, citizen engagement, and data-driven decision-making within the barangay.

Use Case Diagram

The use case diagram used showed the interaction between system users and the system functions.

Figure 2.4 Use Case Diagram



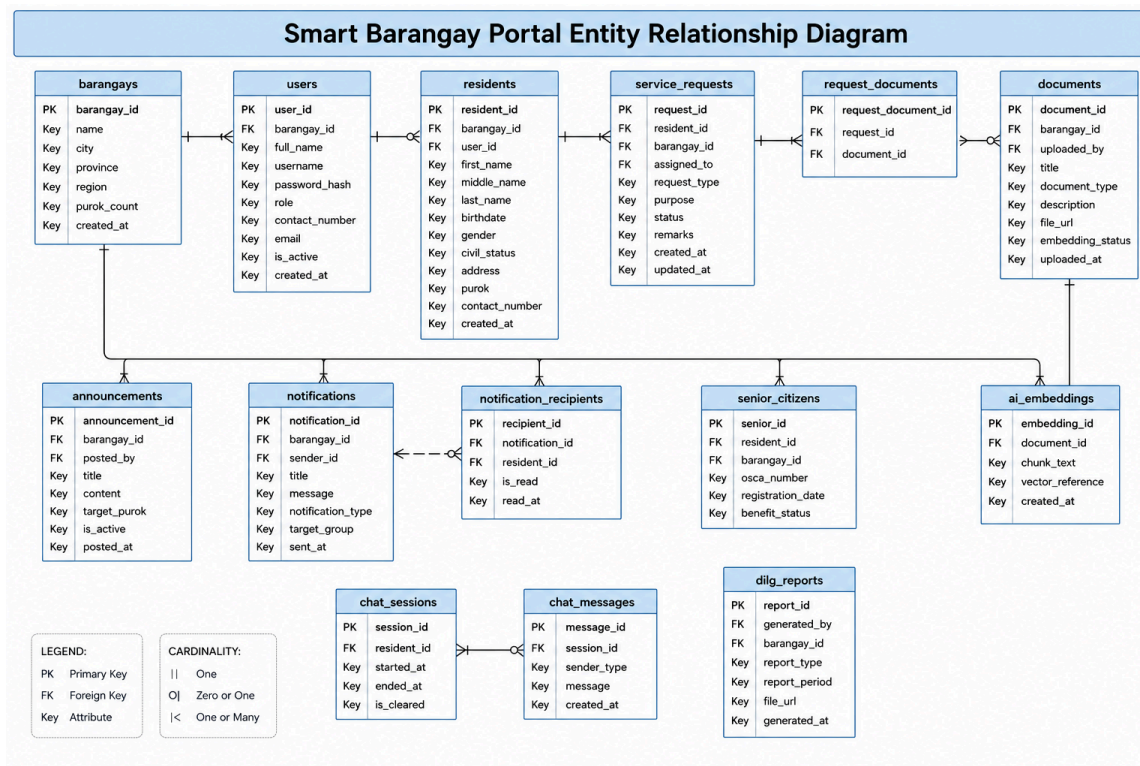


The Use Case Diagram shows the interactions between residents, barangay staff, and the AI-powered system within the Smart Barangay Portal. Residents can submit service requests, communicate with the AI chatbot, view announcements, and track request statuses. Barangay staff manage documents, announcements, senior citizen records, and DILG reports. The AI/System supports automated inquiry handling, information retrieval, and service processing, enabling efficient and accessible barangay services.

Entity Relationship Diagram (ERD)

The ERD defines how the data is structured in the database.

Figure 2.5 Entity Relationship Diagram



The ERD shows how the Smart Barangay Portal organizes and manages its data. Barangays connect to residents, users, service requests, documents, announcements, notifications, senior citizen records. Residents can submit requests, receive notifications, and interact with the AI chatbot, while barangay staff manage records and services. Documents are linked to AI embeddings for

intelligent information retrieval, and chat sessions store conversations between users and the system. This structure ensures efficient data management, accurate record tracking, and reliable delivery of barangay services.

2.2 Software Requirements

2.2.1 Technical Requirements

Component	Requirement
FastApi	Ver
Database	Supabase
Node.js	24.16.0
Framework	LangChain,
Npm	11.17.0

As shown in Table 2.1, these software requirements set the foundation for the system's architecture and ensure compatibility, security, and performance across all components. By adhering to these required versions and platforms, the system is able to provide reliable user authentication,



seamless integration of AI-driven features, and efficient data management throughout all intended operating environments.

Table 2.2 Hardware Requirements

Component	Requirement
Mobile Device	Android 14 and IOS 17+
Desktop	Windows 10+

Environment, Population, and Sample

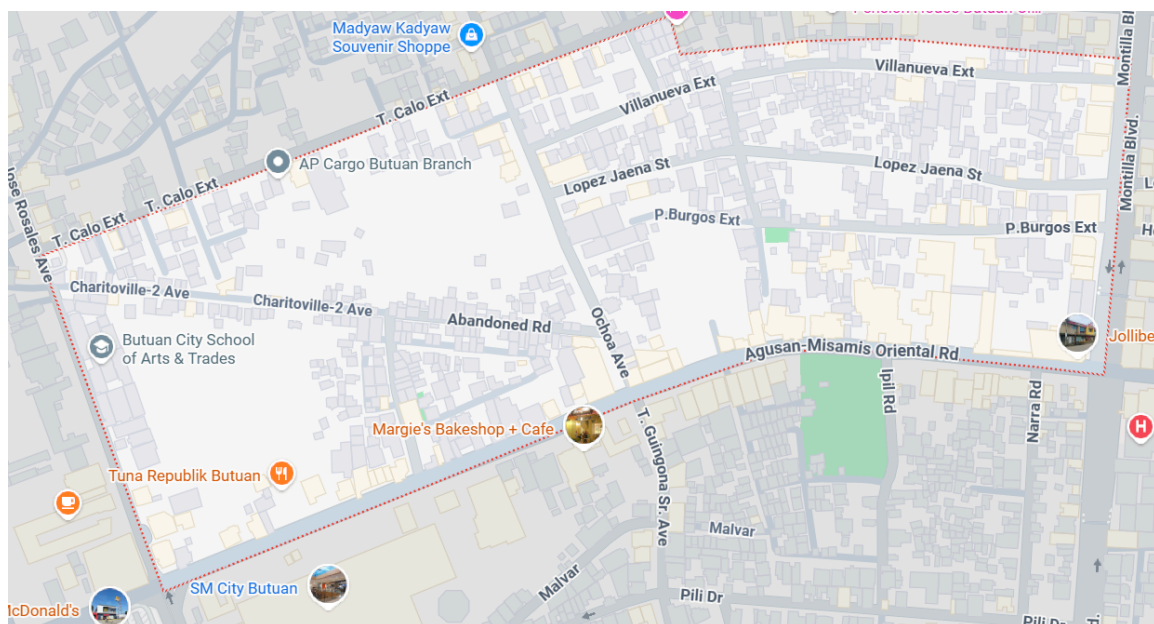
Researched Environment

This study was conducted at **Barangay Tandang Sora (Barangay 12), Butuan City, Agusan del Norte, Caraga Region, Philippines**. Butuan City serves as the regional center of Caraga and has been actively pursuing digital transformation under its Smart Eco City framework, making it a strategically relevant setting for an e-governance study of this nature. Barangay Tandang Sora is a populated urban barangay within Butuan City characterized by a

significant volume of daily resident transactions, including clearance requests, senior citizen registrations all of which are currently managed through manual, paper-based processes.

The system development and testing were carried out using computing facilities at **Father Saturnino Urios University (FSUU), Butuan City**, where the research team developed and hosted the portal in a controlled development environment prior to its pilot deployment in the barangay. The production environment utilized a cloud-based hosting infrastructure compliant with the Philippine Data Privacy Act of 2012, ensuring secure and accessible service delivery for both residents and barangay personnel.

Map of Tandang Sora



Sampling Technique

This study employed **purposive sampling** for the barangay official group, as the 30 participants were deliberately selected based on their designated roles in barangay administration and their direct involvement in the services addressed by the Smart Barangay Portal. All selected officials had functional responsibilities related to at least one of the system's modules document processing, senior citizen registration making them the most appropriate evaluators of the administrative components of the system.

For the resident group, **stratified random sampling** was used to ensure proportional representation across three demographic strata: age group (youth aged 18–35, adults aged 36–59, and senior citizens aged 60 and above), gender (male and female), and technology experience level (low, moderate, and high). This stratification was necessary to test the moderating effects of age, gender, and experience on behavioral intention and use behavior, as specified in the study's UTAUT hypotheses. Within each stratum, participants were selected randomly from the barangay's official household roster to minimize selection bias and ensure that the findings are representative of the broader resident population of Barangay Tandang Sora.



Ethical Consideration

This study adhered strictly to the ethical standards governing research involving human participants, in accordance with the guidelines of Father Saturnino Urios University and the provisions of the **Philippine Data Privacy Act of 2012 (Republic Act No. 10173)**. Prior to data collection, the research team secured the formal endorsement of the **Barangay Captain of Barangay Tandang Sora** and obtained written **informed consent** from all 120 UAT respondents. Each consent form clearly explained the purpose of the study, the nature of their participation, the voluntary nature of their involvement, and their right to withdraw at any time without consequence. No participant was subjected to any form of coercion or undue influence.

Furthermore, the research team committed to transparency in the reporting of all methods, results, and limitations of the study. All sources, frameworks, and prior works referenced in the development and evaluation of the Smart Barangay Portal have been properly cited and acknowledged in accordance with academic integrity standards. No data fabrication, falsification, or plagiarism was committed at any stage of the research process.

Research Instrument

The primary data collection instrument used in this study was a structured **User Acceptance Testing (UAT) questionnaire**, composed of two versions: one for barangay officials and one for residents. Both versions were developed by the research team based on validated scales drawn from the **Technology Acceptance Model (TAM)** and the **SERVQUAL** framework, adapted to the specific context of the Smart Barangay Portal.

The questionnaire was divided into the following parts:

Part I Respondent Profile collected demographic information including age, gender, position (for officials), and self-assessed technology experience level, which served as the moderating variables in the UTAUT analysis.

Part II System Usability and Acceptance contained 20 Likert-scale items (1 = Strongly Disagree to 5 = Strongly Agree) measuring the four UTAUT constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions, as well as Behavioral Intention and Use Behavior.

Part III Service Quality Assessment contained 10 SERVQUAL-based items measuring five service quality dimensions Reliability, Responsiveness,



Assurance, Empathy, and Tangibles as experienced through the portal's interface and AI chatbot.

Part IV Open-ended Feedback provided respondents space to share qualitative observations, suggestions, and concerns about specific system features.

The instrument underwent **content validation** by three expert validators a computer science faculty member, an e-governance practitioner, and a barangay official and was pre-tested on 10 non-sample respondents to establish internal consistency. The Cronbach's alpha reliability coefficient of the final instrument was computed prior to the main data collection, with a target reliability score of $\alpha \geq 0.80$ consistent with the benchmark established by Bringula et al. (2022).

Data Gathering Procedure

The data gathering process for this study was carried out in four sequential phases:

Phase 1 Pre-Development Assessment. Prior to system development,

the research team conducted a needs assessment at Barangay Tandang Sora through direct observation of existing manual processes, informal interviews with barangay officials, and a review of current administrative records. This phase documented the specific inefficiencies in document processing, resident inquiry management, and registry maintenance that the Smart Barangay Portal was designed to address.

Phase 2 System Development. Following the needs assessment, the Smart Barangay Portal was developed in iterative sprints using the waterfall methodology. The research team designed and built the four core modules the AI chatbot, the administrative dashboard, the digital service portal, and the automated reporting system using Langchain for the Development Framework and React/Next.js for the frontend, a RESTful API backend, and a PostgreSQL relational database. The NLP chatbot was trained using a domain-specific corpus of resident inquiry patterns collected from Barangay Tandang Sora, following the Filipino-English BERT architecture outlined by Ramos and Lim (2024). Throughout this phase, interim builds were reviewed by the barangay captain and selected staff members to ensure alignment with actual operational requirements.

Phase 3 Pilot Deployment and UAT. Once the system passed internal



integration testing and quality assurance, it was deployed in a controlled pilot environment accessible to the 120 selected respondents. Barangay officials

were given a structured orientation session on the administrative modules, while residents were given access to the resident-facing portal and chatbot. Each respondent was asked to complete a set of guided tasks such as submitting a document request, using the chatbot to ask about requirements, or generating a sample report before completing the UAT questionnaire. The data collection period for the UAT phase spanned two weeks.

Phase 4 System Performance Benchmarking. Concurrent with the UAT, the research team collected system-level performance data, including chatbot intent recognition accuracy (measured against a set of 200 test queries), report generation time for senior citizen annual reports, and transaction processing speed. These metrics were compared against the pre-established performance targets specified in the study's research objectives.

Prototype

Barangay Sigurado AI

Brgy 12 Tandang Sora

Mag-rehistro

Paliug isulod ang imong impormasyon aron mahimong rehistradong residente.

PANGALAN (FULL NAME)

Juan Dela Cruz

NUMERO SA BARANGAY ID

00-0000-000

NUMERO SA TELEPONO

09XX XXX XXXX

PASSWORD

••••••••

KUMIRMAHA ANG PASSWORD

••••••••

☐ Mouyon ko sa [Terms of Service](#) ug [Privacy Policy](#) sa barangay.

Mag-rehistro

Naa nay account? [Mag-login](#)

SIGURADO & PROTEKTADO



Tandang Sora



Maayong buntag,

Juan dela Cruz

Andam na ba ka motabang sa imong komunidad karong adlaw?



Community Town Hall
Ugma, 2:00 PM sa Barangay Gym.
Importante ang tanan!



Mga Hitabo sa Barangay

Tan-awa Tanan

BAG-O

SABADO



Oplan Linis karong Sabado
7:00 AM - Barangay Hall.
Magtinabangay kita sa paglimpyo sa

Libre
8:00 /
bakur

Iskedyul sa Basura

SMS AKTIBO



Sunod nga Koleksyon
Sabado ug Domingo (Weekend)

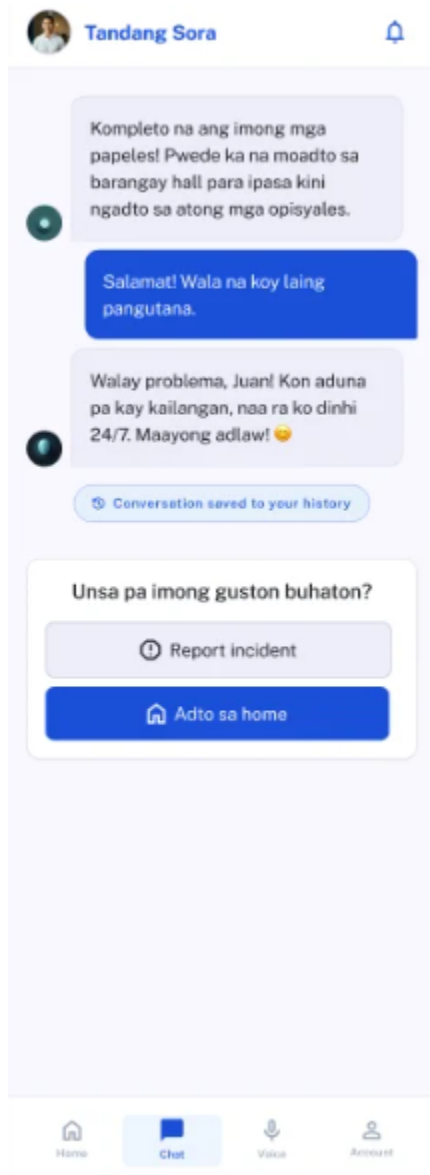
Malata

Dili Malata



AI Warning: Ayaw ilabay ang "Hard Zone" nga basura para sa kaluwesan sa atong mga trabahante.

Paspas nga Serbisyo



**Tandang Sora**

**Juan dela Cruz** VERIFIED
Resident of Barangay 12

 Impormasyon sa Account >

 Mga Notipikasyon >

 Kasaysayan sa Serbisyo >

 Settings sa Pinulongan >

 Tabang ug Suporta >

Log Out

App Version 2.4.1 (Sigurado Build)

**Barangay Sigurado AI**
Brgy 12 Tandang Sora

Mag-login sa inyong account

 Numero sa barangay ID o username

 Password

☐ Hinumdumi ako

Mag-login

o

 Mag-login gamit ang OTP

Wala pay account? Mag-rehistro

Home

Chat

Voice

Account





Barangay Sigurado AI
Brgy 12 Tandang Sora

Mag-login sa inyong account

 Numero sa barangay ID o username

 Password

☐ Hinumdumi ako

Mag-login

 Mag-login gamit ang OTP

[Wala pay account? Mag-rehistro](#)

Barangay Sigurado AI
Brgy 12 Tandang Sora

Mag-rehistro
Palihug isulod ang imong impormasyon aron mahimong rehistradong residente.

PANGALAN (FULL NAME)

 Juan Dela Cruz

NUMERO SA BARANGAY ID

 00-0000-000

NUMERO SA TELEPONO

 09XX XXX XXXX

PASSWORD

 ***** 

KUMPIRMAHA ANG PASSWORD

 *****

☐ Mouyon ko sa [Terms of Service](#) ug [Privacy Policy](#) sa barangay.

Mag-rehistro 

Naa nay account? **Mag-login**

 SIGURADO & PROTEKTADO

**Tandang Sora**

Maayong buntag.

Juan dela Cruz

Andam na ba ka motabang sa imong komunidad karong adlaw?

**Community Town Hall**

Ugma, 2:00 PM sa Barangay Gym.
Importante ang tanan!

Mga Hitabo sa Barangay

Tan-awa Tanan



Oplan Linis karong Sabado
7:00 AM - Barangay Hall.
Magtinabangay kita sa paglimpyo sa

Libre
8:00 /
bakur

Iskedyul sa Basura

SMS AKTIBO

**Sunod nga Koleksyon**
Sabado ug Domingo (Weekend)

☒ Malata

☐ Dili Malata

 **AI Warning:** Ayaw ilabay ang 'Hard Zone' nga basura para sa kaluwesan sa atong mga trabahante.

Paspas nga Serbisyo

**Tandang Sora**

Kompleto na ang imong mga papeles! Pwede ka na moadto sa barangay hall para ipasa kini ngadto sa atong mga opisyales.

Salamat! Wala na koy laing pangutana.

Walay problema, Juan! Kon aduna pa kay kailangan, nsa ra ko dinhi 24/7. Maayong adlaw! 🌞

Conversation saved to your history

Unsa pa imong guston buhaton?

 Report incident

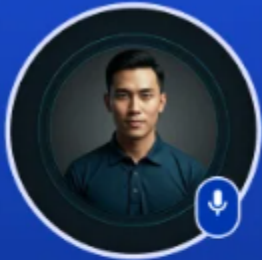
 Adto sa home

 Home

 Chat

 Voice

 Account



IMONG GIINGON:

"Unsay mga r^áquiràmànts
para sa c^àrtificatà of
indig^àncy?"



Undané

Sigurado AI motubag sa Bisaya o Filipino

**Tandang Sora**

**Juan dela Cruz** VERIFIED
Resident of Barangay 12

 Impormasyon sa Account >

 Mga Notipikasyon >

 Kasaysayan sa Serbisyo >

 Settings sa Pinulongan >

 Tabang ug Suporta >

Log Out

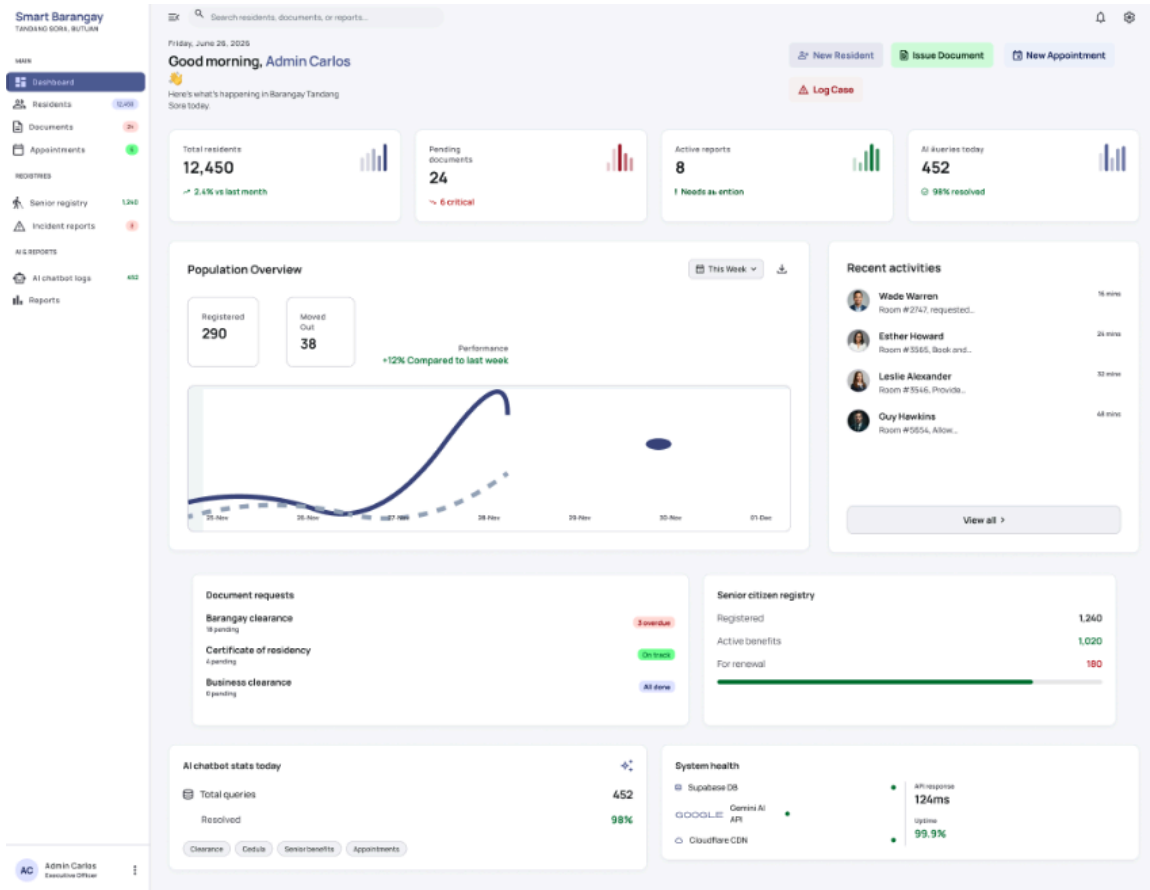
App Version 2.4.1 (Sigurado Build)

Home

Chat

Voice

Account





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SMART BARANGAY

AI-Powered Web and Mobile Services Portal for Barangay Tandang Sora, Butuan City

CAPSTONE THESIS ALIGNMENT REVISION ADDENDUM

Prepared from the submitted 79-page PDF and the current system code audit. Revision date: 31 August 2026. Repository revision audited: 0d23a14.

Decision	Result
Overall alignment	The thesis direction is aligned with the project, but several claims are ahead of the current implementation.
What this revised copy does	Preserves all original thesis pages, then appends corrected wording, an implementation matrix, and acceptance requirements.
Submission warning	Do not claim payment processing, automatic filled-template generation, push/SMS delivery, permanent session deletion, or completed performance results until evidence exists.

How to use this addendum

The original pages are intentionally unchanged because no editable DOCX/ODT source was supplied. For the final defense or submission, use the replacement statements on the following pages in the editable manuscript, then export that manuscript as the official final PDF. This merged PDF is a safe review copy and an auditable record of the required changes.

Basis of review

The review compared the thesis objectives, scope, framework, architecture, ERD, methodology, and requirements against the current resident PWA, admin portal, API, Supabase schema, and registration approval flow. The addendum reports only what can be supported by the current code and database design.

1. Thesis-to-System Alignment Matrix

Status meanings: Implemented = visible in code and reasonably testable; Partial = a working fragment exists but the thesis claim is broader; Missing = no complete implementation or evidence was found.

Thesis / requested capability	Current status	Required correction
Dynamic chatbot form	PARTIAL	The chatbot can trigger a basic request card, but it is not yet a document-type-specific dynamic form with required fields, uploads, validation, and structured storage.
Admin receives requests; one session can contain multiple requests	PARTIAL / BLOCKED	Admin request views exist, but the chat message persistence uses a null session link and request-to-resident linkage must be verified against the residents UUID. Add parent session and child request records.
Ready-for-pickup notification and payment notice	PARTIAL	Notification storage/UI exists, but reliable status-triggered delivery and payment-required data are not complete. Add notification events, payment flag/amount, and delivery/read audit.
Filled template generated from chatbot data	MISSING	Implement a document template registry, field mapping, server-side generation, storage, and staff preview before claiming automation.
Dashboard reports from recorded requests	PARTIAL	Request data and dashboard surfaces exist, but a reproducible report-generation/export flow and tested metrics are required.
Resident registration with staff review/approval	IMPLEMENTED IN CODE; VERIFY DEPLOYMENT	The latest code has pending/approved/rejected flow and RLS migration. Apply migration in the target Supabase project and add applicant notifications and audit evidence.
Clickable questions only when answers are available	PARTIAL	Suggested questions are hard-coded. Build suggestions from approved knowledge records and hide questions with no answer/source.

Important scope correction

The current application is a web/PWA implementation. The thesis should not describe a fully native Android/iOS application unless native builds are actually delivered and tested. The system also has a local knowledge fallback, so RAG answers must be labeled and tested for source provenance.

2. Replacement Statements for the Editable Thesis

Insert these statements into the editable thesis sections. They are written to keep the capstone objective while accurately limiting claims to the current build.

Chapter 1 - Objectives

Replace the claim that the system already automates the complete process from submission through completion and delivery with: *The system supports digital request submission, staff processing, status tracking, and resident notifications for configured services. Automated document generation, payment processing, and final delivery are target extensions unless they have been implemented and validated in the deployed system.*

Chapter 1 - Scope and Delimitations

Use: *The resident interface is delivered as a responsive web/PWA experience. The current chatbot supports knowledge-based answers and a basic request-intake form. Service-specific fields, document uploads, generated templates, payment handling, and external SMS/push delivery are included only for services that have a working end-to-end implementation.*

Conceptual Framework / Architecture

Replace the unconditional dynamic-form and deletion claims with: *The chatbot identifies whether a configured service requires intake data and routes the request to the administrative workflow. Session identity is generated at the client and sent to the API, but persistence, retention, and deletion controls must be explicitly enforced and demonstrated before claiming that conversation data is permanently deleted at interaction end.*

Privacy and Data Protection

Use: *The system applies role-based access and database row-level security in configured environments. Privacy compliance requires a documented retention policy, access audit, deletion job or procedure, consent language, and verification that chat data is isolated by user and session. These controls should be reported as implemented only after deployment testing.*

Methodology and Evaluation

Do not present planned respondent counts, query counts, UAT scores, or performance timings as completed results unless the raw instruments, participants, test runs, and computed results are included. Mark them as proposed evaluation targets until evidence is attached.

3. Required Implementation Before Full Thesis Alignment

These are the minimum acceptance criteria for changing the status from partial/missing to implemented.

Priorit y	Work item	Acceptance evidence
P0	Fix resident identity and request linkage. Use residents.id for document_requests.resident_id, with a reliable auth-user lookup.	A resident submits a request; the same request appears for the correct resident and admin after refresh.
P0	Persist every chatbot message under chat_sessions.id and enforce session ownership.	Database rows contain the session ID; another user cannot read the session; multi-request session test passes.
P0	Define retention/deletion behavior for chat sessions and messages.	Policy, migration/job or documented deletion procedure, and a test showing the expected retention result.
P0	Implement service-specific dynamic forms with schema, required fields, validation, and optional document upload.	At least the defended services produce structured, validated request data.
P1	Add request status events, ready-for-pickup notifications, and payment-required/amount fields.	Approved/released status creates a notification; payment-required cases clearly show amount and next action.
P1	Implement filled-template generation with versioned templates and staff preview.	Generated file contains the submitted resident values and is linked to the request.
P1	Replace hard-coded suggested questions with approved-knowledge availability checks.	Unavailable topics do not appear as clickable questions; each shown question has a traceable answer source.
P1	Deliver dashboard report generation/export from recorded transactions and requests.	Date-filtered counts and downloadable report match database fixtures.
P1	Add applicant approval/rejection notification and complete audit trail.	Applicant sees the result and reason; staff action records actor, time, and decision.

Payment boundary

The thesis requirement says residents should be notified when documents are ready and should be told when payment is required. This does not necessarily require online payment collection. A validated first release may record payment required, amount, payment status, and the in-person payment instruction, provided the thesis clearly says online payment is out of scope.

4. Verification and Defense Evidence

The following evidence should accompany the revised manuscript so that the claims are defensible during capstone review.

Area	Minimum test	Evidence to attach
Registration approval	Register, reject, approve, and log in with each resulting state.	Screenshots, database rows, RLS policy/migration version, and audit log.
Chatbot answers	Ask each clickable question and one unsupported question.	Question inventory, answer source/citation, fallback behavior, and answerability rule.
Multiple requests	Create two document requests in one chat session and one in a new session.	Session/request relational records and admin queue screenshots.
Documents and payment	Process a free service and a payment-required service.	Generated/attached document, payment flag/amount/status, and resident notification.
Reports	Seed known records, filter a date range, export, and reconcile totals.	Exported report plus reconciliation table.
Privacy	Attempt cross-user access and execute the retention/deletion procedure.	Denied-access result, retention configuration, and deletion verification.
Performance/UAT	Run the stated sample sizes only if actually completed.	Raw test data, procedure, date, participants, timings, and computed statistics.

Recommended thesis wording for results

Report measured results separately from target requirements. A feature should be labeled implemented only when the deployed build, database state, and user-visible result agree. If a feature is planned, label it as a limitation or future enhancement rather than presenting it as an achieved output.

Alignment conclusion

The project has a credible foundation for the proposed Smart Barangay system, including resident/admin portals, authentication, an approval workflow in the latest code, a chatbot route, request storage, and dashboard surfaces. Full thesis alignment still depends on completing the end-to-end service workflow and producing the verification evidence listed above.

5. Revision Record and Sign-off

This page records what was changed in this review copy and what remains the responsibility of the development team before submission.

Item	Record
Source reviewed	Copy of the final Barangay AI - IT32.pdf (79 pages), supplied from the Downloads folder.
System revision reviewed	Branch fix/sync-admin-resident-supabase at revision 0d23a14, synchronized with origin/main at the time of audit.
PDF change	Original thesis pages preserved; this six-page alignment addendum appended to the end of the new PDF copy.
Primary correction	Separates implemented, partial, missing, and planned capabilities so the thesis does not overclaim the current build.
Editable-source requirement	For true in-place paragraph replacement, edit the original DOCX/ODT manuscript using the replacement statements in Section 2, then export a new official PDF.

Approval checklist

- ☐ Adviser reviewed the corrected scope and objectives
- ☐ Development team completed or explicitly removed each missing feature claim
- ☐ Database migrations and RLS policies were applied in the target environment
- ☐ End-to-end test evidence is attached to the final manuscript
- ☐ Final thesis was exported from the editable source after these revisions

Prepared for capstone alignment review